Exercise-4

H2C, IOT Project

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Lora

LoRa is a wireless modulation technique derived from Chirp Spread Spectrum (CSS) technology. It is a wireless technology that provides long range, low power and secure data transmission for IoT applications. LoRa Technologies operates in different frequency bands in different regions: In the United States it operates in the 915 MHz band, in Europe it operates in the 868 MHz band and in Asia it operates in the 865 to 867 MHz, 920 to 923 MHz band. (Brian Ray. 2018)

LoRaWAN

LoRaWAN wireless IOT wide-area network *(WAN)* protocols, refer to the open, cloud-based protocol and system architecture for IoT networks that make it possible for LoRa devices to communicate with each other via networks, platforms, and technologies across the Internet. The LoRaWAN ecosystem includes a wide range of device and sensor chip manufacturers, server and base station vendors, and other service providers that are part of the LoRa Alliance. A non-profit association of more than 500 member companies, the LoRa Alliance is arguably the fastest growing technology alliance. The mission of its members is to promote both LPWAN connectivity and IoT ecosystems worldwide through the development and promotion of the LoRaWAN open standard. (Brian Ray. 2018)

NB-IoT

Narrowband IoT or NB-IoT is a wireless communication standard for the Internet of Things in its different forms. NB-IoT belongs to the category of low-power wide-area networks (LPWAN), enabling to connect devices that need small amounts of data, low bandwidth, and long battery life. This makes it suitable for a variety of applications and use cases for IoT. It is a licensed IoT protocol from the 3GPP standard organization. It uses a licensed RF spectrum, which makes it available only through a mobile operator's network. (Brian Ray. 2020)

Comparison

Both LoRaWAN and NB-IoT end devices are architected with lower power consumption in mind. This is achieved by shifting end-devices to sleep mode when they are not in operation. However, LoRaWAN consumes less power than NB-IoT, making it a more viable solution where less maintenance on the end-device is required. Additionally, lower power consumption

makes LoRaWAN a more cost-effective solution than NB-IoT. Since it consumes less power, LoRaWAN also provides a longer battery life (15+ years compared to 10+ years). LoRaWAN is a primary choice for companies providing Energy Harvesting powered IoT solutions. This is one of the main arguments why LoRaWAN can be a better fit for an Industrial IoT application like manufacturing, mining, or oil & gas deployment. Many industrial deployments can be situated in hard to reach remote locations. It would be ideal for such companies if the maintenance of IoT end-devices like sensors could be avoided. For customers searching for a more secure industrial IoT solution, it should be noted that NB-IoT is built on 256-bit 3GPP encryption that is more secure than LoRaWAN and its AES 128-bit encryption. It is always best practice to ensure industrial network security, regardless of connectivity, by using certified devices and working with proven and trusted service providers. (Viezo. 2020)

Currently, NB-IoT and LoRaWAN standards dominate IoT. The network architectures of NB-IoT and LoRaWAN are designed with different applications, power performance needs, and security requirements in mind. For industrial deployments, a network built on LoRaWAN is a more cost-efficient option because we don't have to lease RF spectrum from network operators and because with LoRaWAN, we can find many novel technologies that extend the battery life of sensors. (Viezo. 2020)

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